

Cosmic-Ray Threshold-Rate Reference and Approximation

Scope and data convention

The bundled `data.csv` is interpreted as

$$E \text{ [eV]}, \quad F(E) \text{ [(m}^2 \text{ sr s GeV)}^{-1}\text{]}.$$

The values were digitized from a plotted spectrum rather than supplied as a primary experimental dataset. No point-by-point uncertainties or reproducible original digitization log were included. Consequently, the numerical results below are reference calculations for this CSV, not exact physical predictions, and the displayed precision is intentionally limited. The source graphic is attributed to Sven Lafebre (8 January 2019); this project uses the Creative Commons Attribution-ShareAlike 3.0 option offered on the source file page for the local raster and copyrightable adaptations. See `DATA_PROVENANCE.md` and `NOTICE` for full details.

Explicitly truncated CSV reference

Let $E_{\max}$ be the largest tabulated energy, approximately 3.36×10^{20} eV. For a one-sided horizontal surface of area A exposed to an isotropic downward intensity, the ideal crossing-rate reference used by the code is

$$R_{\text{CSV}}(> E_{\text{th}}; E_{\max}) = \pi A_{\text{m}^2} 10^{-9} \int_{E_{\text{th}}}^{E_{\max}} F(E) dE_{\text{eV}}.$$

For $A = 1 \text{ km}^2 = 10^6 \text{ m}^2$ this becomes

$$R_{\text{CSV}}(> E_{\text{th}}; E_{\max}) = \pi \times 10^{-3} \int_{E_{\text{th}}}^{E_{\max}} F(E) dE_{\text{eV}}.$$

The upper limit is finite by definition. The calculation does *not* assert that the physical flux is zero above $E_{\max}$.

The code also offers an explicit **last-segment** option. If the final log-log segment is extrapolated as $F(E) = F(E_{\max})(E/E_{\max})^{\gamma_{\text{last}}}$ with $\gamma_{\text{last}} < -1$, its model-dependent tail contribution is

$$\int_{E_{\max}}^{\infty} F(E) dE = \frac{F(E_{\max})E_{\max}}{-(\gamma_{\text{last}} + 1)}.$$

This extrapolation can materially change results near the top of the CSV and is therefore never applied silently.

Acceptance assumptions

The factor πA is the projected geometric acceptance of an ideal, one-sided, horizontal surface for a downward isotropic intensity. It is not a complete instrument response. Detector efficiency, dead time, trigger and reconstruction response, atmospheric shower development, and energy- or direction-dependent effective area are not included. The Python interface can instead accept a caller-supplied scalar aperture in $\text{m}^2 \text{ sr}$, but realistic instruments usually require integrating a full energy- and direction-dependent response.

Piecewise approximation

The following approximation is calibrated to the bundled CSV, the finite $E_{\max}$ truncation convention, a flat horizontal 1 km^2 acceptance, and the range

$$10^{10} \text{ eV} \leq E \leq 10^{20} \text{ eV}.$$

Its units are $\text{s}^{-1} \text{ km}^{-2}$. The coefficients and validation statistics are generated by the companion Python script from the same implementation used by the command-line program.

$$R_{\text{fit}}(> E) \approx \begin{cases} 4.061 \times 10^8 \left(\frac{E}{10^{10} \text{ eV}} \right)^{-1.5452}, & 10^{10} \leq E < 10^{12} \text{ eV}, \\ 3.298 \times 10^5 \left(\frac{E}{10^{12} \text{ eV}} \right)^{-1.6679}, & 10^{12} \leq E < 2 \times 10^{15} \text{ eV}, \\ 1.029 \left(\frac{E}{2 \times 10^{15} \text{ eV}} \right)^{-1.9606}, & 2 \times 10^{15} \leq E < 5 \times 10^{16} \text{ eV}, \\ 1.87 \times 10^{-3} \left(\frac{E}{5 \times 10^{16} \text{ eV}} \right)^{-2.2102}, & 5 \times 10^{16} \leq E < 3 \times 10^{18} \text{ eV}, \\ 2.196 \times 10^{-7} \left(\frac{E}{3 \times 10^{18} \text{ eV}} \right)^{-1.9198}, & 3 \times 10^{18} \leq E < 3 \times 10^{19} \text{ eV}, \\ 2.642 \times 10^{-9} \left(\frac{E}{3 \times 10^{19} \text{ eV}} \right)^{-2.6838}, & 3 \times 10^{19} \leq E \leq 10^{20} \text{ eV}, \end{cases}$$

The fit was sampled at 10,001 thresholds uniformly spaced in $\log_{10} E$. The resulting absolute relative errors were median 5.43%, 95th percentile 16.06%, and worst sampled value 18.44%. These are sampled statistics, not global mathematical bounds.

Selected thresholds

The table uses integer-decade thresholds and modest output precision. The ‘‘CSV-truncated reference’’ column integrates only to the final tabulated point. Values near 10^{20} eV should therefore be treated as sensitive to the unspecified high-energy tail.

Threshold E_{th} [eV]	Piecewise fit [$\text{s}^{-1} \text{ km}^{-2}$]	CSV-truncated reference [$\text{s}^{-1} \text{ km}^{-2}$]
10^{12}	3.298×10^5	3.298×10^5
10^{13}	7.085×10^3	6.031×10^3
10^{14}	1.522×10^2	1.519×10^2
10^{15}	3.271	3.474
10^{16}	4.387×10^{-2}	4.953×10^{-2}
10^{17}	4.04×10^{-4}	4.332×10^{-4}
10^{18}	2.49×10^{-6}	2.406×10^{-6}
10^{19}	2.177×10^{-8}	2.225×10^{-8}
10^{20}	1.044×10^{-10}	1.044×10^{-10}

For an ideal flat surface of another area A in km^2 , both columns scale linearly with A . The documented fit is not automatically applied to modified CSV data or another tail convention; it must be explicitly recalibrated.